

Natural language statement

Domain

GroundTruth

V.13.1. The Eberlein-Smulian Theorem. If X is a Banach space and A subset X , then the following statements are equivalent.
 (a) Each sequence of elements of A has a subsequence that is weakly convergent. (b) Each sequence of elements of A has a weak cluster point. (c) The weak closure of A is weakly compact.

Functional analysis

```
theorem conway_V_13_1_eberlein_smulian
  {E : Type*} [NormedAddCommGroup E] [NormedSpace C E] [CompleteSpace E]
  (A : Set E) :
  let subsequences := forall u : N -> E, (forall n, u n in A) ->
    exists phi : N -> N, StrictMono phi /\ exists x : E,
      Tendsto (fun n => toWeakSpace C E (u (phi n))) atTop (nhds
        (toWeakSpace C E x))
  let clusterPoints := forall u : N -> E, (forall n, u n in A) ->
    exists x : E, MapClusterPt (toWeakSpace C E x) atTop (fun n =>
      toWeakSpace C E (u n))
  List.TFAE [subsequences, clusterPoints,
    IsCompact (closure (toWeakSpace C E '' A))] := by
  sorry
```

V.13.3. James's Theorem. If X is a Banach space and A is a closed convex subset of X such that for each x^* in X^* there is an x_0 in A with $|\langle x_0, x^* \rangle| = \sup\{|\langle x, x^* \rangle| : x \in A\}$, then A is weakly compact.

Functional analysis

```
theorem conway_V_13_3_james
  {E : Type*} [NormedAddCommGroup E] [NormedSpace C E] [CompleteSpace E]
  (A : Set E) (hAclosed : IsClosed A) (hAconvex : Convex R A)
  (hattains : forall phi : E -> L[C] C,
    exists x0 in A, forall x in A, ||phi x|| <= ||phi x0||) :
  IsCompact (toWeakSpace C E '' A) := by
  sorry
```

VI.2.1. The Banach-Stone Theorem. If X and Y are compact and $T : C(X) \rightarrow C(Y)$ is a surjective isometry, then there is a homeomorphism $\tau : Y \rightarrow X$ and a function α in $C(Y)$ such that $|\alpha(y)| = 1$ for all y and $(Tf)(y) = \alpha(y)f(\tau(y))$ for all f in $C(X)$ and y in Y .

Functional analysis

```
theorem conway_VI_2_1_banach_stone
  {X Y : Type*} [TopologicalSpace X] [TopologicalSpace Y]
  [CompactSpace X] [CompactSpace Y] [T2Space X] [T2Space Y]
  (T : (X -> C) -> L[C] (Y -> C)) (hT : Function.Surjective T) :
  exists tau : Y equivt X, exists alpha : Y -> C,
    (forall y : Y, ||alpha y|| = 1) /\ forall f : X -> C, forall y : Y,
      T f y = alpha y * f (tau y) := by
  sorry
```

VIII.3.6. Theorem. If A is a C^* -algebra and a in A , then the following statements are equivalent. (a) $a \geq 0$. (b) $a = b^2$ for some b in $\text{Re } A$. (c) $a = x^*x$ for some x in A . (d) $a = a^*$ and $||t - a|| \leq t$ for all $t \geq ||a||$. (e) $a = a^*$ and $||t - a|| \leq t$ for some $t \geq ||a||$.

Operator algebras

```
theorem conway_VIII_3_6_positive_element_characterizations
  {A : Type*} [CStarAlgebra A] [PartialOrder A] [StarOrderedRing A] (a :
  A) :
  let hermitianSquare := exists b : A, IsSelfAdjoint b /\ a = b ^ 2
  let starSquare := exists x : A, a = star x * x
  let normBoundForAll := IsSelfAdjoint a /\ forall t : R, ||a|| <= t ->
    ||algebraMap C A (t : C) - a|| <= t
  let normBoundForSome := IsSelfAdjoint a /\ exists t : R, ||a|| <= t /\
    ||algebraMap C A (t : C) - a|| <= t
  List.TFAE [0 <= a, hermitianSquare, starSquare, normBoundForAll,
    normBoundForSome] := by
  sorry
```

VII.7.1. Theorem. (F. Riesz) If $\dim X = \infty$ and A in $B_0(X)$, then one and only one of the following possibilities occurs.
 (a) $\sigma(A) = \{0\}$. (b) $\sigma(A) = \{0, \lambda_{bd1}, \dots, \lambda_{bdn}\}$, where for $1 \leq k \leq n$, $\lambda_{bdk} \neq 0$, each λ_{bdk} is an eigenvalue of A , and $\dim \ker(A - \lambda_{bdk}) < \infty$. (c) $\sigma(A) = \{0, \lambda_{bda1}, \lambda_{bda2}, \dots\}$, where for each $k \geq 1$, λ_{bdak} is an eigenvalue of A , $\dim \ker(A - \lambda_{bdak}) < \infty$, and, moreover, $\lim \lambda_{bdak} = 0$.

Operator theory

```
theorem conway_VII_7_1_riesz_compact_operator_spectrum
  {E : Type*} [NormedAddCommGroup E] [NormedSpace C E] [CompleteSpace E]
  (hE : not FiniteDimensional C E) (T : E -> L[C] E) (hT :
  IsCompactOperator T) :
  spectrum C T = {0} \ /
  (exists n : N, 0 < n /\ exists eig : Fin n -> C,
    Function.Injective eig /\ (forall i, eig i != 0) /\
    spectrum C T = insert 0 (range eig) /\
    forall i, (exists x : E, x != 0 /\ T x = eig i * x) /\
      FiniteDimensional C
      (LinearMap.ker
        (T - eig i * ContinuousLinearMap.id C E).toLinearMap)) \ /
  exists eig : N -> C,
    Function.Injective eig /\ (forall n, eig n != 0) /\
    spectrum C T = insert 0 (range eig) /\ Tendsto eig atTop (nhds 0) /\
    forall n, (exists x : E, x != 0 /\ T x = eig n * x) /\
      FiniteDimensional C
      (LinearMap.ker
        (T - eig n * ContinuousLinearMap.id C E).toLinearMap)) := by
  sorry
```

VIII.5.17. Theorem. If A is a C^* -algebra, then there is a representation (π, H) of A such that π is an isometry. If A is separable, then H can be chosen separable.

Operator algebras

```
theorem conway_VIII_5_17_gelfand_naimark {A : Type u} [CStarAlgebra A] :
  (exists (H : Type u) ( _ : NormedAddCommGroup H) ( _ : InnerProductSpace
  C H)
    ( _ : CompleteSpace H) (pi : A -> a[C] (H -> L[C] H)), Isometry pi) /\
  (TopologicalSpace.SeparableSpace A ->
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		<pre> exists (H : Type u) (_ : NormedAddCommGroup H) (_ : InnerProductSpace C H) (_ : CompleteSpace H) (_ : TopologicalSpace.SeparableSpace H) (pi : A ->a[C] (H ->L[C] H)), Isometry pi) := by sorry </pre>
Definition. A bounded operator $A : H \rightarrow H$ is left semi-Fredholm if it is left invertible modulo the compact operators: there are bounded $B : H \rightarrow H$ and compact $C : H \rightarrow H$ such that $BA = 1 + C$. XI.2.3. Theorem. If $A : H \rightarrow H$ is a bounded operator, the following statements are equivalent. (a) A is left semi-Fredholm. (b) $\text{ran } A$ is closed and $\dim \ker A < \infty$. (c) There is a bounded operator $B : H \rightarrow H$ and a finite rank operator F on H such that $BA = 1 + F$. (d) There is no sequence $\{h_n\}$ of unit vectors in H such that $h_n \rightarrow 0$ weakly and $\lim \ Ah_n\ = 0$. (e) There is no orthonormal sequence $\{e_n\}$ in H such that $\lim \ Ae_n\ = 0$. (f) There is a $\delta > 0$ such that $\{h \in H : \ Ah\ \leq \delta\}$ contains no infinite dimensional manifold. (g) If the positive operator $(A^*A)^{1/2} = \int_0^\infty f(t) dE(t)$, then there is a $\delta > 0$ such that $E[0, \delta]H$ is finite dimensional. (h) If $K \in B_0(H)$, then $\dim \ker(A + K) < \infty$.	Fredholm theory	<pre> structure ProjectionValuedMeasure (H : Type*) [NormedAddCommGroup H] [InnerProductSpace C H] [CompleteSpace H] where toFun : Set C -> H ->L[C] H empty : toFun empty = 0 univ : toFun univ = ContinuousLinearMap.id C H projection : forall B : Set C, MeasurableSet B -> IsOrthogonalProjection (toFun B) nonmeasurable : forall B : Set C, not MeasurableSet B -> toFun B = 0 orthogonal : forall B C : Set C, MeasurableSet B -> MeasurableSet C -> Disjoint B C -> (toFun B).comp (toFun C) = 0 countablyAdditive : forall (B : N -> Set C), (forall n, MeasurableSet (B n)) -> Pairwise (fun m n => Disjoint (B m) (B n)) -> forall x : H, Tendsto (fun N => sum n in Finset.range N, toFun (B n) x) atTop (nhds (toFun (union n, B n) x)) def IsLeftSemiFredholm {H K : Type*} [NormedAddCommGroup H] [NormedAddCommGroup K] [InnerProductSpace C H] [InnerProductSpace C K] [CompleteSpace H] [CompleteSpace K] (A : H ->L[C] K) : Prop := exists B : K ->L[C] H, exists C : H ->L[C] H, IsCompactOperator C /\ B.comp A = ContinuousLinearMap.id C H + C theorem conway_XI_2_3_left_semi_fredholm_characterizations {H K : Type*} [NormedAddCommGroup H] [NormedAddCommGroup K] [InnerProductSpace C H] [InnerProductSpace C K] [CompleteSpace H] [CompleteSpace K] (A : H ->L[C] K) : let a := IsLeftSemiFredholm A let b := IsClosed (range A) /\ FiniteDimensional C (LinearMap.ker A.toLinearMap) let c := exists B : K ->L[C] H, exists F : H ->L[C] H, FiniteDimensional C (LinearMap.range F.toLinearMap) /\ B.comp A = ContinuousLinearMap.id C H + F let d := not exists u : N -> H, (forall n, u n = 1) /\ Tendsto (fun n => toWeakSpace C H (u n)) atTop (nhds 0) /\ Tendsto (fun n => A (u n)) atTop (nhds 0) let e := not exists u : N -> H, Orthonormal C u /\ Tendsto (fun n => A (u n)) atTop (nhds 0) let f := exists delta : R, 0 < delta /\ forall M : Submodule C H, (forall x : M, A x <= delta * (x : H)) -> FiniteDimensional C M let g := exists modulus : H ->L[C] H, modulus.comp modulus = A.adjoint.comp A /\ modulus.adjoint = modulus /\ (forall x : H, 0 <= (inner C (modulus x) x).re) /\ exists E : ProjectionValuedMeasure H, E.toFun (spectrum C modulus) = ContinuousLinearMap.id C H /\ (exists scalarMeasure : H -> H -> ComplexMeasure C, (forall x y : H, forall B : Set C, MeasurableSet B -> scalarMeasure x y B = inner C (E.toFun B x) y) /\ forall x y : H, inner C (modulus x) y = integralv z, z d[ContinuousLinearMap.mul R C; scalarMeasure x y]) /\ exists delta : R, 0 < delta /\ FiniteDimensional C (LinearMap.range (E.toFun (Metric.closedBall (0 : C) delta)).toLinearMap) let h := forall C : H ->L[C] K, IsCompactOperator C -> FiniteDimensional C (LinearMap.ker (A + C).toLinearMap) List.TFAE [a, b, c, d, e, f, g, h] := by sorry </pre>
II.7.6. Spectral Theorem for Compact Normal Operators. If T is a compact normal operator on the complex Hilbert space H, then T has only a countable number of distinct eigenvalues. If	Operator theory	<pre> abbrev IsOrthogonalProjection {H : Type*} [NormedAddCommGroup H] [InnerProductSpace C H] [CompleteSpace H] (P : H ->L[C] H) : Prop := IsStarProjection P </pre>

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$\{\lambda_1, \lambda_2, \dots\}$ are the distinct nonzero eigenvalues of T , and P_n is the projection of H onto $\ker(T - \lambda_n)$, then $P_n P_m = P_m P_n = 0$ if $n \neq m$ and $T = \sum_{n=1}^{\infty} \lambda_n P_n$, where this series converges to T in the metric defined by the norm on $B(H)$.

```
theorem conway_II_7_6_compact_normal_spectral_theorem
{H : Type*} [NormedAddCommGroup H] [InnerProductSpace C H]
[CompleteSpace H]
(T : H → L[C] H) (hnormal : T.adjoint.comp T = T.comp T.adjoint)
(hcompact : IsCompactOperator T) :
exists (iota : Type) (ι : Countable iota) (eigenvalue : iota → C)
(projection : iota → H → L[C] H),
(forall i, eigenvalue i != 0) /\ Function.Injective eigenvalue /\
(forall i, projection i != 0) /\ (forall i, IsOrthogonalProjection
(projection i)) /\
Pairwise (fun i j => (projection i).comp (projection j) = 0) /\
(forall i, LinearMap.range (projection i).toLinearMap =
LinearMap.ker (T - eigenvalue i * ContinuousLinearMap.id C
H).toLinearMap) /\
(forall eps : ℝ, 0 < eps → {i : iota | eps ≤ ||eigenvalue
i||}.Finite) /\
HasSum (fun i => eigenvalue i * projection i) T := by
sorry
```

IX.2.2. The Spectral Theorem. If N is a normal operator, there is a unique spectral measure E on the Borel subsets of $\sigma(N)$ such that: (a) $N = \int z dE(z)$; (b) if G is a nonempty relatively open subset of $\sigma(N)$, $E(G) \neq 0$; (c) if A in $B(H)$, then $AN = NA$ and $AN^* = N^*A$ if and only if $AE(\Delta)A = E(\Delta)A$ for every Δ .

Operator
theory

```
abbrev IsOrthogonalProjection {H : Type*} [NormedAddCommGroup H]
[InnerProductSpace C H] [CompleteSpace H] (P : H → L[C] H) : Prop :=
IsStarProjection P

structure ProjectionValuedMeasure (H : Type*) [NormedAddCommGroup H]
[InnerProductSpace C H] [CompleteSpace H] where
toFun : Set C → H → L[C] H
empty : toFun empty = 0
univ : toFun univ = ContinuousLinearMap.id C H
projection : forall B : Set C, MeasurableSet B → IsOrthogonalProjection
(toFun B)
nonmeasurable : forall B : Set C, not MeasurableSet B → toFun B = 0
orthogonal : forall B C : Set C, MeasurableSet B → MeasurableSet C →
Disjoint B C →
(toFun B).comp (toFun C) = 0
countablyAdditive : forall (B : N → Set C), (forall n, MeasurableSet (B
n)) →
Pairwise (fun m n => Disjoint (B m) (B n)) → forall x : H,
Tendsto (fun N => sum n in Finset.range N, toFun (B n) x) atTop
(nhds (toFun (union n, B n) x))

theorem conway_IX_2_2_bounded_normal_spectral_theorem
{H : Type*} [NormedAddCommGroup H] [InnerProductSpace C H]
[CompleteSpace H]
(T : H → L[C] H) (hnormal : T.adjoint.comp T = T.comp T.adjoint) :
exists! E : ProjectionValuedMeasure H,
E.toFun (spectrum C T) = ContinuousLinearMap.id C H /\
(exists scalarMeasure : H → H → ComplexMeasure C,
(forall x y : H, forall B : Set C, MeasurableSet B →
scalarMeasure x y B = inner C (E.toFun B x) y) /\
forall x y : H, inner C (T x) y =
integralv z, z d(ContinuousLinearMap.mul R C; scalarMeasure x y))
/\
(forall G : Set C, G.Nonempty →
(exists 0 : Set C, IsOpen 0 /\ G = 0 inter spectrum C T) → E.toFun
G != 0) /\
forall A : H → L[C] H,
(A.comp T = T.comp A /\ A.comp T.adjoint = T.adjoint.comp A) <=>
forall Delta : Set C, MeasurableSet Delta → A.comp (E.toFun
Delta) = (E.toFun Delta).comp A := by
sorry
```

X.5.6. Stone's Theorem. If U is a strongly continuous one parameter unitary group, then there is a self-adjoint operator A such that $U(t) = \exp(itA)$.

Operator
theory

```
structure ProjectionValuedMeasure (H : Type*) [NormedAddCommGroup H]
[InnerProductSpace C H] [CompleteSpace H] where
toFun : Set C → H → L[C] H
empty : toFun empty = 0
univ : toFun univ = ContinuousLinearMap.id C H
projection : forall B : Set C, MeasurableSet B → IsOrthogonalProjection
(toFun B)
nonmeasurable : forall B : Set C, not MeasurableSet B → toFun B = 0
orthogonal : forall B C : Set C, MeasurableSet B → MeasurableSet C →
```

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Disjoint B C ->
  (toFun B).comp (toFun C) = 0
countablyAdditive : forall (B : N -> Set C), (forall n, MeasurableSet (B
n)) ->
  Pairwise (fun m n => Disjoint (B m) (B n)) -> forall x : H,
  Tendsto (fun N => sum n in Finset.range N, toFun (B n) x) atTop
  (nhds (toFun (union n, B n) x))

structure DenselyDefinedOperator (H : Type*) [NormedAddCommGroup H]
  [InnerProductSpace C H] where
  domain : Submodule C H
  op : domain ->L[C] H
  dense_domain : Dense (domain : Set H)

def IsSelfAdjointUnbounded {H : Type*} [NormedAddCommGroup H]
  [InnerProductSpace C H] (A : DenselyDefinedOperator H) : Prop :=
  (forall y : H, y in A.domain <->
    exists z : H, forall x : A.domain, inner C (A.op x) y = inner C (x : H)
    z) /\
  forall y : A.domain, forall z : H,
    (forall x : A.domain, inner C (A.op x) y = inner C (x : H) z) -> z =
    A.op y

def IsUnitaryOperator {H : Type*} [NormedAddCommGroup H]
  [InnerProductSpace C H] [CompleteSpace H] (U : H ->L[C] H) : Prop :=
  U.adjoint.comp U = ContinuousLinearMap.id C H /\
  U.comp U.adjoint = ContinuousLinearMap.id C H

def StronglyContinuousUnitaryGroup {H : Type*} [NormedAddCommGroup H]
  [InnerProductSpace C H] [CompleteSpace H] (U : R -> H ->L[C] H) : Prop
  :=
  U 0 = ContinuousLinearMap.id C H /\
  (forall s t : R, U (s + t) = (U s).comp (U t)) /\
  (forall t : R, IsUnitaryOperator (U t)) /\
  forall x : H, Continuous fun t => U t x

def IsSpectralExponential {H : Type*} [NormedAddCommGroup H]
  [InnerProductSpace C H]
  [CompleteSpace H] (A : DenselyDefinedOperator H) (U : R -> H ->L[C] H)
  : Prop :=
  exists (E : ProjectionValuedMeasure H) (scalarMeasure : H -> H ->
  ComplexMeasure C),
  E.toFun {z : C | z.im != 0} = 0 /\
  (forall x y : H, forall B : Set C, MeasurableSet B ->
    scalarMeasure x y B = inner C (E.toFun B x) y) /\
  (forall x : A.domain, forall y : H, inner C (A.op x) y =
    integralv z, z d[ContinuousLinearMap.mul R C; scalarMeasure x y]) /\
  forall t : R, forall x y : H, inner C (U t x) y =
    integralv z, Complex.exp (Complex.I * t * z)
    d[ContinuousLinearMap.mul R C; scalarMeasure x y]

theorem conway_X_5_6_stone_theorem
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace C H]
  [CompleteSpace H]
  (U : R -> H ->L[C] H) (hU : StronglyContinuousUnitaryGroup U) :
  exists! A : DenselyDefinedOperator H,
    IsSelfAdjointUnbounded A /\ IsSpectralExponential A U := by
  sorry

```